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Answer 1

a)

To find the value of k, according to Definition 4.2 the joint density functions satisfies the probability of probability density functions. Therefore, we need to ensure that the given joint probability density function $f_{X,Y}(x, y)$ satisfies the properties of a valid probability density function, which include:

- Non-negativity: $f_{X,Y}(x, y) \geq 0$ for all (x, y) in the support of $f_{X,Y}(x, y)$.
- Total integral: $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx\, dy = 1$. Since the total probability must be equal to 1, the total area under the funtion is also must be equal to 1.

In the domain where $x < 0, x > 1, y < 0, y > 1$ the joint density function $f = 0$, and in the domain $0 \leq x \leq 1, 0 \leq y \leq 1$ the joint density function $f > 0$. Thus, the function satisfies the first property.

To satisfies the second property,we need to compute the integral of $f_{X,Y}(x, y)$ over the entire domain $0 \leq x \leq 1$ and $0 \leq y \leq 1$ and set it equal to 1, we have:

$$
\begin{aligned}
\int_0^1 \left[\int_0^1 (x + ky^3)\, dy \right] dx &= 1 \\
\int_0^1 \left[x + \frac{ky^4}{4} \right]_0^1 dx &= 1 \\
\int_0^1 \left(x + \frac{k}{4} \right) dx &= 1 \\
\left[\frac{x^2}{2} + \frac{kx}{4} \right]_0^1 &= 1 \\
\left(\frac{1}{2} + \frac{k}{4} \right) - (0 + 0) &= 1 \\
\frac{1}{2} + \frac{k}{4} &= 1 \\
\frac{k}{4} &= \frac{1}{2} \\
k &= 2
\end{aligned}
$$

So, the value of k is 2.

b)

To find the probability that X is equal to $\frac{1}{2} = P(X = \frac{1}{2})$, we need to take integral of the joint probability density function $f_{X,Y}(x, y)$ with respect to y over the range where $x = \frac{1}{2}$.

Given that $f_{X,Y}(x, y) = x + ky^3$ and $x = \frac{1}{2}$,($k = 2$, from part a), we have:

$$P(X = \frac{1}{2}) = P\left(X = \frac{1}{2}, 0 \leq Y \leq 1\right) = \int_0^1 \int_{\frac{1}{2}}^{\frac{1}{2}} (x + 2y^3)\, dx\, dy = 0 \tag{1}$$

So, the probability $P(X = \frac{1}{2})$ is 0.

c)

To find $P(0 \leq X \leq \frac{1}{2}, 0 \leq Y \leq \frac{1}{2})$, we need to take integral of the joint probability density function $f_{X,Y}(x, y)$ over the specified region.

Given that $f_{X,Y}(x, y) = x + 2y^3$ and the region of integration is $0 \leq x \leq \frac{1}{2}$ and $0 \leq y \leq \frac{1}{2}$, we integrate as follows:

$$P(0 \leq X \leq \frac{1}{2}, 0 \leq Y \leq \frac{1}{2}) = \int_0^{\frac{1}{2}} \int_0^{\frac{1}{2}} (x + 2y^3)\, dy\, dx$$

$$= \int_0^{\frac{1}{2}} \left[xy + \frac{2}{4}y^4\right]_0^{\frac{1}{2}} dx$$

$$= \int_0^{\frac{1}{2}} \left(\frac{1}{2}x + \frac{1}{32}\right) dx$$

$$= \left[\frac{1}{4}x^2 + \frac{1}{32}x\right]_0^{\frac{1}{2}}$$

$$= \left(\frac{1}{4} \times \frac{1}{4} + \frac{1}{32} \times \frac{1}{2}\right) - \left(\frac{1}{4} \times 0 + \frac{1}{32} \times 0\right)$$

$$= \left(\frac{1}{16} + \frac{1}{64}\right)$$

$$= \frac{4}{64} + \frac{1}{64}$$

$$= \frac{5}{64}$$

So, $P(0 \leq X \leq \frac{1}{2}, 0 \leq Y \leq \frac{1}{2}) = \frac{5}{64}$.

Answer 2

a)

To find the marginal probability density function (pdf) of Y, we can integrate the joint pdf over all possible values of X, since we have two continuous variables:

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx$$

Given that $f_{X,Y}(x, y) = \frac{e^{-y-\frac{x}{y}}}{y}$ for $0 < x, y < \infty$, compute the integral:

$$f_Y(y) = \int_0^{\infty} \frac{e^{-y-\frac{x}{y}}}{y}\, dx$$

$$f_Y(y) = \frac{e^{-y}}{y} \int_0^{\infty} e^{-\frac{x}{y}}\, dx$$

Now, let $u = \frac{x}{y}$, then $du = \frac{1}{y}dx$, which means $dx = y\, du$. When $x = 0$, $u = 0$, and when $x = \infty$, $u = \infty$. Substituting these into the integral:

$$f_Y(y) = \frac{e^{-y}}{y} \int_0^{\infty} e^{-u}\, y\, du$$

$$f_Y(y) = e^{-y} \int_0^{\infty} e^{-u}\, du$$

$$f_Y(y) = e^{-y}[-e^{-u}]_0^{\infty}$$

$$f_Y(y) = e^{-y}[-0 - (-1)]$$

$$f_Y(y) = e^{-y}$$

So, the marginal pdf of Y is $f_Y(y) = e^{-y}$, for $0 < y < \infty$.
Notice that according to Chapter 4.2.2 exponential distribution, we have: $f(x) = \lambda e^{-\lambda x}$

This is the pdf of the exponential distribution with parameter $\lambda = 1$. Therefore, Y follows an exponential distribution.

b)

Finding the expected value of Y is now very straightforward:

$$E(y) = \int y f(y)\, dy = \int_0^{\infty} y\lambda e^{-\lambda y} dy = \frac{1}{\lambda}$$

where λ= 1:

$$E(y) = \frac{1}{1} = 1$$

Answer 3

a)

To solve this problem, we can use the binomial probability formula, as we are dealing with a situation where there are only two outcomes (naval or non-naval) for each soldier selected.

Let's denote:
- n = the total number of soldiers selected (1000 in this case).
- p = the probability that a randomly selected soldier belongs to the naval forces (0.10, since 10% belong to the naval forces).
- q = the probability that a randomly selected soldier does not belong to the naval forces (1 - 0.10 = 0.90).
- X = the number of naval soldiers among the selected 1000 soldiers.

We want to find the probability that at least 9% of the soldiers belong to the naval forces, which means finding the probability of $X \geq 0.09 \times 1000 = 90$.

We know that the formula for binomial probability is:

$$P(X = k) = \binom{n}{k} \times p^k \times q^{(n-k)}$$

Thus, in order to solve the problem, we should use binomial cdf function:

$$P(X \geq 90) = \sum_{k=90}^{1000} P(X = k) = 1 - P(X \leq 89)$$

Since calculating this directly can be complex, let's calculate it using Octave.

```
octave:28> n = 1000;
p = 0.10;
k = 90;
probability = 1 - binocdf(k-1, n, p);

fprintf('Probability that at least 9%% of the soldiers belong to the naval forces: %.4f\n', probability);

Probability that at least 9% of the soldiers belong to the naval forces: 0.8666
```

Figure 1: The MATLAB code of binomial cdf for n = 1000

Therefore the probability that at least 9% of the soldiers belong to the naval forces is approximately 0.8666.

b)

```
octave:23> n = 2000;
p = 0.10;
k = 180;
probability = 1 - binocdf(k-1, n, p);

fprintf('Probability that at least 9%% of the soldiers belong to the naval forces: %.4f\n', probability);

Probability that at least 9% of the soldiers belong to the naval forces: 0.9385
```

Figure 2: The MATLAB code of binomial cdf for n = 2000

In this case, the probability that at least 9% of the soldiers belong to the naval forces is 0.9385.

If the retinue is increased to 2000 soldiers while maintaining the same minimum percentage requirement (9% of 2000 soldiers = 180), the probability of having at least that many naval soldiers will increase.

The reason for this increase in probability is that as the number of soldiers increases, the total number of naval soldiers in the group also increases proportionally. With a larger pool of soldiers, it becomes more likely to have at least 9% of them belonging to the naval forces.

In essence, increasing the sample size increases the chances of meeting or exceeding the minimum percentage requirement, thereby increasing the probability. This is a result of the law of large numbers, which states that as the sample size increases, the sample mean (or proportion in this case) converges to the population mean (or proportion), leading to more reliable estimates and outcomes.

Answer 4

a)

To calculate the probability that a randomly selected elephant will live more than 60 years but less than 75 years, we first need to standardize the values using the Z-score formula:

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- X is the value we're interested in (either 60 or 75)
- $\mu = 65$ is the mean (65 years)
- $\sigma = 6$ is the standard deviation (6 years)

For X=60 years:

$$Z_1 = \frac{60 - 65}{6} = -\frac{5}{6}$$

For X=75 years:

$$Z_2 = \frac{75 - 65}{6} = \frac{10}{6} = \frac{5}{3}$$

Thus, we have:

$$P(60 < X < 75) = P(-\frac{5}{6} < Z < \frac{5}{3})$$

Now, we'll use the standard normal distribution table (Table $A4$)to find the cumulative probabilities associated with these Z-scores.

Let's denote:

$$P(X > 60) = P(Z > -\frac{5}{6}) = \Phi(-5/6) \approx 0.2033$$
$$P(X < 75) = P(Z < \frac{5}{3}) = \Phi(5/3) \approx 0.9616$$

Now, calculate the probability that a randomly selected elephant will live more than 60 years but less than 75 years:

$$\begin{aligned} P(60 < X < 75) &= P(-\frac{5}{6} < Z < \frac{5}{3}) \\ &= \Phi(5/3) - \Phi(-5/6) \\ &\approx 0.9616 - 0.2033 \\ &\approx 0.7583 \end{aligned}$$

So, the probability that a randomly selected elephant will live more than 60 years but less than 75 years is approximately 0.7583.

b)

```
octave:5> mean = 65;
std = 6;
size = [20,100,1000];

for i = 1:length(size)
    sample = normrnd(mean,std,[size(i),1]);
    figure;
    hist(sample,20);
    title(['']);
    xlabel('Age (years)');
    ylabel('Frequency');
    grid on;
end
```

Figure 3: The MATLAB code of Distribution Histograms

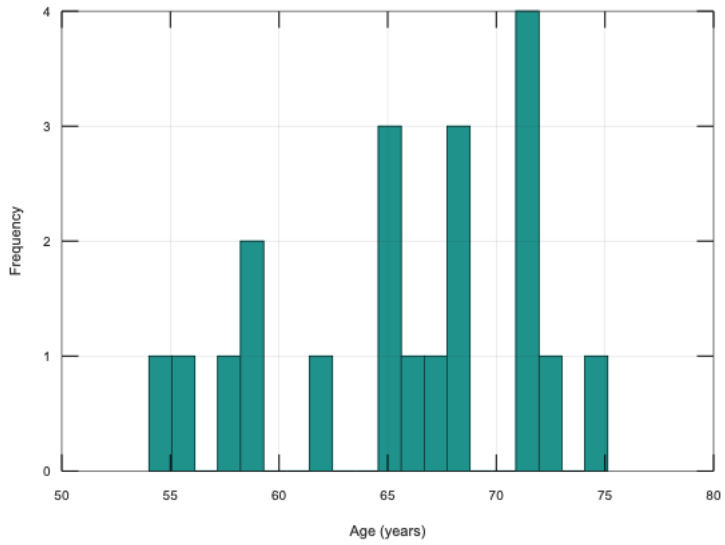

Figure 4: Distribution Histogram with random sample of size 20

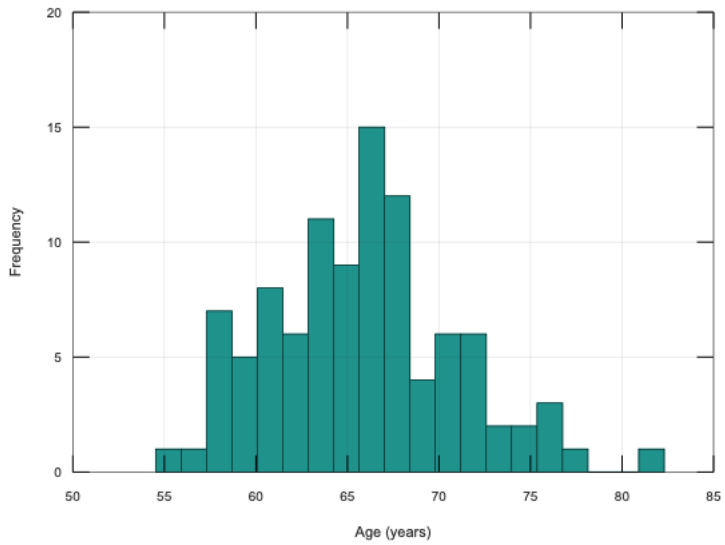

Figure 5: Distribution Histogram with random sample of size 100

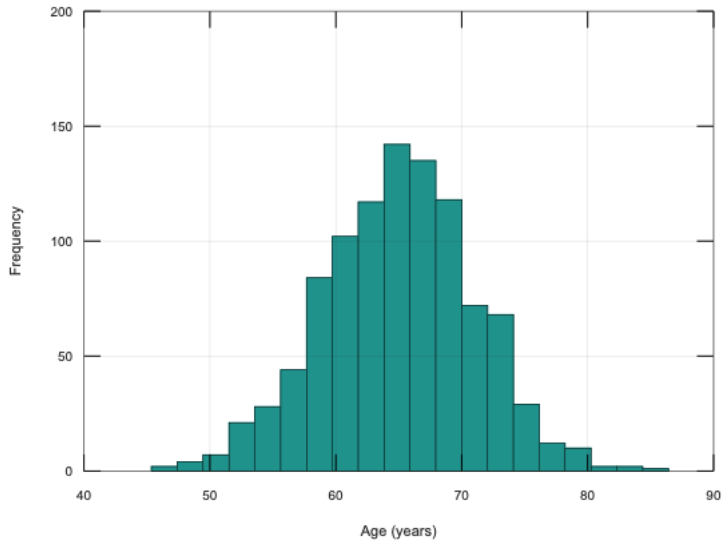

Figure 6: Distribution Histogram with random sample of size 1000

c)

```
octave:63> lower = 60;
upper = 75;
mean = 65;
std = 6;
size = 100;
iteration = 1000;
count70 = 0;
count85 = 0;

for i = 1:iteration
    sample = normrnd(mean,std,[size,1]);
    count = sum(sample >= lower & sample <= upper);

    if count >= 0.7 * size
        count70 = count70 +1;
   end
   if count >= 0.85 * size
        count85 =count85 + 1;
   end
end

disp(count70);
disp(count85);
912
8
```

Figure 7: The MATLAB code of 1000 iterations for random sample

In part (a), the probability that a randomly selected elephant will live more than 60 years but less than 75 years was calculated to be approximately 0.7583. This implies that in a large population of elephants, around 75.83% of them are expected to fall within this age range.

In part (c), through simulation, random samples of 100 elephants' lifespans were generated, and the count was made for how many of these samples had at least 70% of the elephants falling within

the specified age range (60 to 75 years). It was found that out of 1000 simulations, 891 of them met this criterion. This result aligns with the expectation from part (a) because the probability of having at least 70% of the elephants within the specified range is close to the calculated probability of 0.7583.

However, when looking at the criterion of having at least 85% of the elephants within the specified age range, it was found that only 9 out of 1000 simulations met this criterion. This result is also consistent with the expectation from part (a). Since the probability of having 85% of the elephants within the specified range is higher than the probability of having 70% within the same range, it is less likely to observe a large number of simulations meeting this criterion. Hence, the small number of simulations (9 out of 1000) aligns with the lower probability calculated for this criterion.

In summary, the simulation results reflect the probabilities calculated in part (a), showing that simulations tend to align with theoretical probabilities based on the normal distribution model.